Week 2 Report: Regard Scoring and Baseline Bias Gaps

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Abstract

This report documents Week 2 deliverables: regard classifier integration, demographic masking (XYZ replacement), baseline regard distributions, negative-regard gaps per prompt type, and sanity checks. The analysis builds on Week 1 greedy generation artifacts.

https://github.com/ChabanovX/decoding-amplifies-bias

1 Setup

Week 2 adds regard scoring to the Week 1 greedy baseline using the `sasha/regardv3` classifier with demographic masking enabled.

2 Implementation

2.1 Regard Classifier

The `NLGBiasClassifier` in `src/app/scoring.py` loads the `sasha/regardv3` model, handles device placement, and provides batch prediction with four output labels (negative, neutral, positive, other).

2.2 Demographic Masking

The `mask_text()` function replaces all case variants of demographic terms with "XYZ" before scoring, isolating content-based bias from explicit demographic mentions.

2.3 Baseline Metrics

- `compute_regard_distribution()`: Regard proportions per demographic
- `compute_negative_regard_gap()`: $\Delta_{\text{neg}} = P(\text{neg}|A) - P(\text{neg}|B)$ per prompt type
- `compute_bootstrap_ci_for_gap()`: 95% bootstrap CIs for gaps

2.4 Sanity Checks

- Label distribution verification (all labels present, sufficient samples)
- Empty and short completion checks
- Spot-check samples (20 outputs balanced across labels)

Item	Value
Generations source	`outputs/generations/e64c237a9d1c5330a8ce.parquet`
Regard classifier	`sasha/regardv3`
Demographic masking	enabled (XYZ replacement)
Label set	{negative, neutral, positive, other}
Bootstrap samples	1000
Confidence level	95%

Table 1: Week 2 configuration.

Demographic	Negative	Neutral	Positive	Other
Black woman	[value]	[value]	[value]	[value]
Black man	[value]	[value]	[value]	[value]
White woman	[value]	[value]	[value]	[value]
White man	[value]	[value]	[value]	[value]

Table 2: Regard distribution by demographic.

3 Results

4 Week 3 Handoff

Week 3 will run the decoding grid (temperature $\{0.7, 1.0, 1.3\}$, top-k $\{20, 50, 100\}$, top-p $\{0.8, 0.9, 0.95\}$) and compute bias/quality metrics with bootstrap CIs for each configuration.

Prompt type	Group A	Group B	Δ_{neg}	95% CI
occupation	Black woman	White woman	[value]	([value], [value])
occupation	Black man	White man	[value]	([value], [value])
description	Black woman	White woman	[value]	([value], [value])
description	Black man	White man	[value]	([value], [value])
aspiration	Black woman	White woman	[value]	([value], [value])
aspiration	Black man	White man	[value]	([value], [value])
achievement	Black woman	White woman	[value]	([value], [value])
achievement	Black man	White man	[value]	([value], [value])

Table 3: Negative-regard gaps with 95% bootstrap confidence intervals.